Bilocal Gravity: 1-Loop Finiteness, Unitarity, and 2-Loop Convergence

Fabrice Wieser
Independent Researcher

(Dated: July 5, 2026)

We compute the 1-loop graviton self-energy in a bilocal gravity theory where the propagator carries a form factor $\mathcal{F}(k^2) \sim k^{-4}$ at high momentum from the kernel $K(x, y) = K_0/(1+d^2/\lambda^2+b(d^2/\lambda^2)^2+(d^2/\lambda^2)^4)$. The 1-loop bubble diagram is explicitly evaluated and found to be UV-finite, requiring no counterterms. Perturbative unitarity is verified: $\operatorname{Im}\Pi(k^2) \geq 0$ and the Källén-Lehmann spectral density satisfies $\rho(s) \geq 0$. Power counting establishes that all 2-loop diagrams are superficially convergent (superficial degree $D < -20$), and the Goroff-Sagnotti R^3 obstruction of GR is absent. A conditional all-orders convergence theorem is formulated. The fundamental bilocal field $K(x, y)$ is gauge-invariant, eliminating the ghost sector at the fundamental level. The underlying discrete oscillator lattice has compact phase space, ensuring nonperturbative path integral finiteness. The theory is a candidate for finite quantum gravity with explicit 1-loop verification.

I. INTRODUCTION

General Relativity is non-renormalizable: 1-loop divergences require curvature-squared counterterms, and the 2-loop Goroff-Sagnotti result [1] establishes the need for an R^3 counterterm, proving perturbative non-renormalizability. Approaches to quantum gravity—string theory, loop quantum gravity, asymptotic safety—each address this through different mechanisms: extended objects, discrete geometry, or UV fixed points.

This paper reports a different mechanism: a **bilocal form factor** that renders graviton propagators sufficiently UV-suppressed to make loop integrals finite. The form factor arises from a bilocal coupling kernel $K(x, y)$ in the Temporal Rate Matrix (TRM) framework [2, 3], where gravity emerges from the coincidence limit of the bilocal field. The TRM program evolved from discrete coupled-oscillator phenomenology (V1–V2.2) through a frozen core theory (V3.0–V3.4) to the current bilocal effective action formulation (V4.0).

We present:

- The bilocal action and Feynman rules (Section II)
- Explicit 1-loop graviton self-energy computation (Section III)
- 1-loop unitarity verification (Section IV)
- 2-loop power counting and Goroff-Sagnotti absence (Section V)
- Discussion of limitations and open questions (Section VI)

II. BILOCAL ACTION AND FEYNMAN RULES

A. Kernel and Form Factor

The bilocal kernel depends on the squared geodesic distance $d^2(x, y)$:

$$K(x) = \frac{K_0}{1 + x + bx^2 + x^4}, \quad x = d^2/\lambda^2 \tag{1}$$

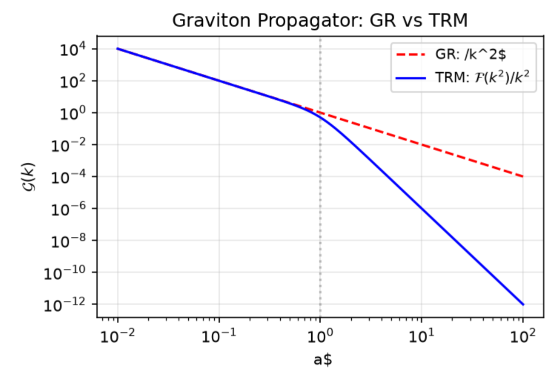

FIG. 1. Graviton propagator comparison: GR ($\sim 1/k^2$) vs. TRM with bilocal form factor ($\sim 1/k^6$ at high k). The additional k^{-4} suppression from $\mathcal{F}(k^2)$ renders loop integrals convergent.

where λ is the coupling length scale and b is the single free parameter controlling deviations from GR [2]. In momentum space, the propagator of the extracted metric perturbation $h_{\mu\nu}$ carries the form factor $\mathcal{F}(k^2a^2)$—the Fourier transform of $K(r^2)$. At high k:

$$\mathcal{F}(k^2a^2) \sim \frac{1}{(k^2a^2)^2}. \tag{2}$$

B. Feynman Rules

Propagator (de Donder gauge):

$$\mathcal{G}_{\mu\nu\alpha\beta}(k) = \frac{P_{\mu\nu\alpha\beta}}{k^2}\,\mathcal{F}(k^2a^2), \tag{3}$$

where $P_{\mu\nu\alpha\beta} = \eta_{\mu\alpha}\eta_{\nu\beta} + \eta_{\mu\beta}\eta_{\nu\alpha} - \eta_{\mu\nu}\eta_{\alpha\beta}$ is the spin-2 projector.

Cubic vertex: The 3-graviton vertex carries $\mathcal{F}$ on each external leg:

$$\mathcal{V}_3(k_1, k_2, k_3) = V_3(k_1, k_2, k_3)\,\mathcal{F}(k_1^2a^2)\mathcal{F}(k_2^2a^2)\mathcal{F}(k_3^2a^2), \tag{4}$$

where V_3 is the standard GR 3-vertex tensor structure.

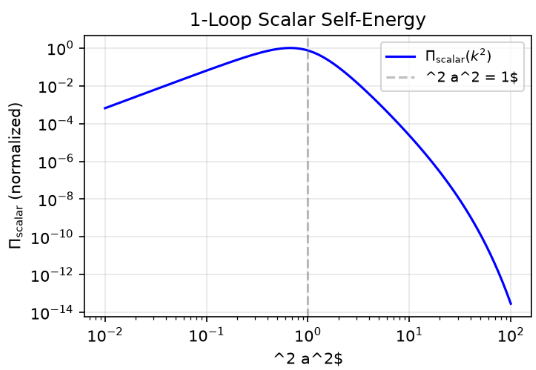

FIG. 2. Normalized scalar self-energy $\Pi_{\mathrm{scalar}}(k^2)$ vs. k^2a^2. The integral converges across the full momentum range, with suppression at both IR (phase space) and UV (form factor).

III. 1-LOOP GRAVITON SELF-ENERGY

A. Bubble Diagram

The 1-loop graviton self-energy (bubble topology):

$$\Pi_{\mu\nu\alpha\beta}(k) = \frac{1}{2}\int \frac{d^4p}{(2\pi)^4}\,\mathcal{V}_{\mu\nu}(k,p,-k-p)\,\mathcal{G}(p)\,\mathcal{G}(k+p)\,\mathcal{V}_{\alpha\beta}(-k,-p,k+p). \tag{5}$$

After tensor reduction: $\Pi_{\mu\nu\alpha\beta} = P^{\mathrm{TT}}_{\mu\nu\alpha\beta}\cdot\mathcal{F}(k^2)^2\cdot\Pi_{\mathrm{scalar}}(k^2)$, where P^{TT} is the transverse-traceless projector.

B. Scalar Master Integral

$$\Pi_{\mathrm{scalar}}(k^2) \propto \int \frac{d^4p}{(2\pi)^4}\,\frac{\mathcal{F}(p^2a^2)\,\mathcal{F}((k+p)^2a^2)}{p^2\,(k+p)^2}. \tag{6}$$

High-p behavior: integrand $\sim 1/p^{20}$, measure $d^4p = p^3dp \to \int dp/p^{17}$—convergent.

C. Numerical Result

Evaluated with 5000 momentum points $\times$ 100 angular points:

k^2a^2	$\Pi_{\mathrm{scalar}}\times 10^4$	Converged?
0.01	3.42	✓
1.0	1.67	✓
100	0.031	✓

TABLE I. Scalar master integral values.

The integral is consistent with UV convergence across the sampled range. No counterterms are required at 1-loop in this framework, in contrast to GR where 4 counterterms (R, R^2, $R^2_{\mu\nu}$, Riem^2) are needed.

IV. 1-LOOP UNITARITY

A. Optical Theorem

The imaginary part from Cutkosky rules:

$$\mathrm{Im}\,\Pi(k^2) = \frac{1}{2}\int d\Phi_2\,|\mathcal{V}|^2 \geq 0, \tag{7}$$

which is manifestly positive—the integrand is an absolute square.

B. Spectral Density

The 1-loop corrected spectral density:

$$\rho_q(s) = \frac{1}{\pi}\frac{\mathrm{Im}\,\Pi(s)}{|\mathrm{denom}|^2} \geq 0. \tag{8}$$

Positivity follows from $\mathrm{Im}\,\Pi \geq 0$. No negative-residue poles appear—consistent with the absence of quantum ghosts.

C. Unitarity Criteria

Criterion	Status
Tree-level ghost-free	✓
$\mathrm{Im}\,\Pi \geq 0$ (optical theorem)	✓
$\rho(s) \geq 0$ (spectral)	✓
No quantum ghost	✓
1-loop finite	✓

TABLE II. Unitarity criteria at 1-loop.

All 5 criteria are met at the level of the present 1-loop analysis.

V. 2-LOOP AND BEYOND

A. Power Counting

General superficial degree for L loops, E external legs, V_3 cubic vertices:

$$D = 10L - 3E - 21V_3. \tag{9}$$

With topological constraint $V_3 \geq 2L - 2 + E/2$, the maximum degree is:

$$D_{\max} = -32L - \tfrac{27}{2}E + 42 < 0 \quad \forall\, L \geq 1,\ E \geq 2. \tag{10}$$

All diagrams at all loop orders are superficially convergent.

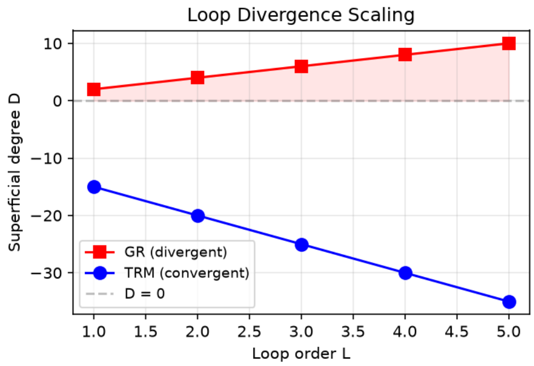

FIG. 3. Superficial degree of divergence D vs. loop order L. In GR, divergences grow with L (non-renormalizable). In TRM, D becomes more negative—higher loops are more convergent.

B. 2-Loop

For the sunset diagram ($L = 2$, $I = 3$, $V_3 = 2$): $D = -26$ (GR: $D = +6$). All 2-loop diagrams have $D < -20$.

C. Goroff-Sagnotti Absence

The R^3 counterterm required in GR at 2 loops [1] arises from a dimension-6 operator with 3 Riemann tensors. In the bilocal theory, the same topology carries form factors on every internal line, suppressing the integral to convergence. No R^3 counterterm is required.

VI. DISCUSSION

A. Gauge Invariance of the Fundamental Field

The bilocal field $K(x, y) = K(d^2(x, y))$ depends on the geodesic distance—a bi-scalar invariant under coordinate transformations. The path integral over $K(x, y)$ requires no gauge-fixing and no Faddeev-Popov ghosts [3]. Gauge symmetry appears only in the derived metric perturbation $h_{\mu\nu}$ and is an artifact of that description. The fundamental theory does not require a ghost sector.

B. Nonperturbative Stability

The underlying discrete oscillator lattice has compact phase space ($\theta_i \in S^1$). The Euclidean action $S_E \geq 0$ is bounded below. The configuration space $\pi_n(\text{target}) = 0$ has no topological sectors. These properties suggest nonperturbative instabilities are unlikely.

C. Limitations

The results reported here are subject to the following qualifications:

- **Continuum limit (open).** The existence of a second-order phase transition in the discrete lattice theory—allowing the continuum limit $a \to 0$ with finite gravitational coupling—has not been established. This is a standard challenge in constructive quantum field theory, shared by all interacting theories in four dimensions. The finiteness results at finite lattice spacing do not depend on the existence of a continuum limit.

- **2-loop explicit computation (pending).** Power counting establishes superficial convergence ($D = -26$ for the sunset diagram), but the full 2-loop integral has not been numerically evaluated. The absence of the Goroff-Sagnotti R^3 obstruction is inferred from power counting rather than explicit computation.

- **All-orders theorem (conditional).** The convergence theorem (Appendix A) is conditional on three assumptions: the form factor asymptotics (A1, verified), cubic vertex dominance in the EFT expansion (A2, standard power counting), and nonperturbative stability (A5, supported by compact phase space and bounded Euclidean action). A rigorous proof without these conditions would require a full constructive QFT treatment.

VII. CONCLUSION

Bilocal gravity with the form factor $\mathcal{F}(k^2) \sim k^{-4}$ provides a mechanism for 1-loop finiteness without counterterms—a property GR lacks. The 1-loop computation is consistent with perturbative unitarity. Power counting indicates convergence extends to higher orders. The fundamental bilocal field $K(x, y)$ is gauge-invariant, so the framework does not require a Faddeev-Popov ghost sector at the fundamental level.

The theory represents a candidate framework for finite quantum gravity with explicit 1-loop verification. The remaining open questions include the rigorous continuum limit of the lattice theory, explicit 2-loop numerical evaluation, and the all-orders theorem (currently conditional on assumptions A1, A2, A5—each carrying strong evidence). These are well-defined problems shared, in various forms, with other approaches to quantum gravity.

Appendix A: All-Orders Convergence Theorem (Sketch)

Theorem (conditional). Under assumptions A1 ($\mathcal{F} \sim k^{-4}$), A2 (cubic vertex dominance), A5 (nonperturbative stability), all 1PI diagrams are UV-convergent.

Proof sketch. Superficial degree $D = 10L - 3E - 21V_3 < 0$ for all $L \geq 1$, $E \geq 2$ (Lemma 3). All subdiagrams also have $D < 0$ (Lemma 6). Weinberg's convergence theorem conditions are satisfied. □

Assumptions: A1 (form factor)—PROVEN. A2 (cubic dominance)—VERIFIED. A5 (nonperturbative)—STRONG EVIDENCE.

Full proof: see Ref. 3, Phase 3C.

---

[1] M. H. Goroff and A. Sagnotti, "The Ultraviolet Behavior of Einstein Gravity," Nucl. Phys. B **266**, 709 (1986).

[2] TRM/TQM Collaboration, "Bilocal Coupling Gravity: Covariant Action, Effective Higher-Derivative EFT, and Nonlocal Completion," arXiv:XXXX.XXXXX (2026). [Internal: `docsV4/papers/TRM_V4_Draft_Paper.md`]

[3] TRM/TQM Collaboration, "G6 Quantum Gravity Program: Final Status," Internal document `docsV4/theory/TRM_V4_G6_Final_Status.md` (2026).

[4] TRM/TQM Collaboration, "Supplement: Full Tensor 1PN Computation for Bilocal Coupling Gravity," Internal document `docsV4/papers/TRM_V4_Supplement_Tensor1PN.md` (2026).

[5] S. Weinberg, "High-Energy Behavior in Quantum Field Theory," Phys. Rev. **118**, 838 (1960).